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A principal component analysis of the *U.S. News & World Report* tier rankings of colleges and universities

Thomas J. Webster *

Lubin School of Business, Department of Finance & Economics, Pace University, One Pace Plaza, New York, NY 10038, USA

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Abstract

This paper utilizes principal component regression analysis to examine the relative contributions of 11 ranking criteria used to construct the *U.S. News & World Report* (USNWR) tier rankings of national universities. The main finding of this study is that the actual contributions of the 11 ranking criteria examined differ substantially from the explicit USNWR weighting scheme because of severe and pervasive multicollinearity among the ranking criteria. USNWR assigns the greatest weight to academic reputation. However, generated first principal component eigenvalues of tier rankings indicate that the most significant ranking criterion is the average SAT scores of enrolled students. This result is significant since admission requirements are policy variables that indirectly affect, for example, admission applications, yields, enrollment, retention, tuition-based revenues, and alumni contributions.

JEL classification: A22; I29

Keywords: Principal component analysis; *U.S. News & World Report*; Colleges and universities; Rankings

1. Introduction

In recent years the *U.S. News & World Report* (USNWR) tier rankings of colleges and universities, which range from tier 1 (highest) to tier 4 (lowest), have become an important source of information for and about colleges and universities. USNWR tier rankings are important to prospective students since this information makes the search process more efficient and less costly. The tier rankings are important for college and university administrators because they partly define the institution's market niche, influence the perception of the institution by prospective students, which affects enrollments and operating budgets, and serve as a guide to the institution's strategic planning. In fact, senior college and university administrators regularly meet with USNWR editors to explore the reasons for their institutions' decline in tier rankings. The USNWR tier rankings are also important to college recruiters who use the rankings to allocate recruitment budgets. Finally, the importance of the USNWR tier rankings is likely to increase because of their ready availability on the internet.

The USNWR tier rankings are the subject of much debate and controversy due to their growing significance to students and administrators, or perhaps because of it. Much of this controversy revolves around the apparently arbitrary weighting scheme applied to the ranking criteria. The USNWR tier rankings of colleges and universities affect crucial operational aspects of ranked institutions (see, for example Carter, 1998; Crissey, 1997; Garigliano, 1997; Gilley, 1992; Glass, 1997; Gleick, 1995; Graham & Diamond, 1999; Kirk & Corcoran, 1995; Machung, 1998; Morse & Gilbert, 1995; Schatz, 1993). These rankings influence the number and quality of admission applications, which affect the overall character of the institution's student body, the quality of

* Tel.: +1-212-346-1965; fax: +1-212-346-1976.
E-mail address: twebster@pace.edu (T.J. Webster).

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the institution's programs, and, ultimately, the perceived value of the institution's degree. These perceptions are likely to influence retention rates, in particular, and enrollment measures, in general. Enrollment, in turn, affects the institution's tuition-based revenues and, therefore, financial resources, operating budgets, per student expenditures, faculty/student ratios, etc.

These factors, in turn, are likely to further influence the academic reputation of the college or university, which could impact alumni contributions, foundation grants, and other non-tuition-based revenue sources. This would suggest that important feedback effects are imbedded in the ranking process where tier rankings reinforce existing positive and negative stereotypes about the academic quality of ranked institutions. To be sure, the USNWR tier rankings are but one of a myriad of factors that influence the institutional decision-making process. Nevertheless, it also seems fair to say that the USNWR tier rankings of colleges and universities directly and indirectly influence a number of short-run and long-run aspects of college and university operations.

2. Ranking criteria and weights

U.S. News & World Report bases its college and university rankings on a set of up to 16 measures of academic quality that fall into seven broad categories: academic reputation, student selectivity, faculty resources, student retention, financial resources, alumni giving, and, for national universities and national liberal arts colleges only, graduate rate 'performance'. USNWR ranking criteria and weights are summarized in Table 1.

An examination of the ranking criteria used by USNWR suggests that multicollinearity is pervasive. Multicollinearity in this instance refers to the degree to which changes in the value of one or more of the ranking criteria are related to, and are affected by, changes in one or more of the other ranking criteria. It could be argued, for example, that an institution's academic reputation is influenced by knowledge of the SAT scores of admitted students. Retention rates, enrollments, and alumni contributions are likely to be affected by academic reputation, which, in turn, would influence an institution's financial resources, per-student expenditures, faculty/student ratios, faculty compensations, etc.

The possibility of pervasive multicollinearity between and among the ranking criteria suggests that the assigned weighting scheme may not accurately reflect actual contributions to USNWR tier rankings. The purpose of this paper is to analyze the accuracy of the USNWR criteria weighting scheme and to discuss the implications of the findings for the formulation of marketing strategies of college and university administrators.

3. Data

Data used in this study were obtained from the website of *U.S. News & World Report 1999 College Rankings*. USNWR categorizes institutions of higher learning as either national universities, national liberal arts colleges, regional universities, or regional liberal arts colleges according to criteria established by the Carnegie Foundation for the Advancement of Teaching. There were 228 national universities included in the USNWR tier rankings. According to USNWR, a national university offers a full range of undergraduate majors. It also offers master's and doctoral degrees.

A national liberal arts college, on the other hand, emphasizes undergraduate education. To be classified as a national liberal arts college, at least 40 percent of conferred degrees must be in the liberal arts. These institutions tend to require higher college entrance examination scores than regional liberal arts colleges. There were 162 national liberal arts colleges in the USNWR rankings.

Regional universities and liberal arts colleges are subdivided into four regions: Midwest, North, South, and West. Regional universities offer a full range of undergraduate programs and master's degree programs. Unlike national universities, however, regional universities generally do not confer doctoral degrees. There are 504 regional universities in the 1998 USNWR rankings.

Regional liberal arts colleges emphasize undergraduate education and tend to be somewhat less selective than national liberal arts colleges. There are 423 regional liberal arts colleges in the USNWR rankings. However, regional liberal arts colleges grant fewer than 40 percent of their degrees in liberal arts. In general, each category (a national university, national liberal arts college, regional university, and regional liberal arts college) is divided into four, approximately equal sized, tiers that are based on a variety of weighted ranking criteria.

This study is restricted to national universities reporting SAT scores. This restriction is justified on the grounds that the present study should be viewed as a first step in a comprehensive regional and categorical analysis of USNWR tier rankings of colleges and universities. On the other hand, this restriction is potentially problematic since it may introduce a regional bias into the analysis as a greater proportion of colleges and universities in the Midwest, West, and South accept ACT test scores than do colleges and universities in the East.

Average SAT scores utilized in this study were calculated by averaging USNWR reported first and fourth quartile average SAT scores. There were 145 national universities, or about 64 percent of all national universities, included in the tier rankings, that reported SAT scores. In general, the higher the tier ranking, the greater the percentage of ranked institutions that require SAT scores for admission.

Table 1
U.S. News & World Report. Undergraduate ranking criteria and weights

Criterion	Definition
Academic reputation	This indicator is the average rating of the quality of the school's academic programs as evaluated by officials at similar institutions. The survey was conducted in the spring of 1998 (25 percent)
Acceptance rate	This is the ratio of the number of students admitted to the number of applicants for admission for the fall 1997 semester (2.25 percent)
Alumni giving	The percentage of undergraduate alumni of record who donated money to the institution during the years 1996 and 1997 (5 percent)
Class size, 1–19 students	The percentage of undergraduate classes, excluding class sections, with fewer than 20 students enrolled during the fall 1997 semester (6 percent)
Class size, 50+ students	The percentage of undergraduate classes, excluding class sections, with 50 students or more enrolled during the fall 1997 semester (2 percent)
Expenditure per student	Total educational expenditures per full-time-equivalent student (10 percent)
Faculty compensation	Average faculty pay and benefits adjusted for regional differences in cost of living during the 1996 and 1997 academic years. Includes full-time assistant, associate, and full professors (7 percent)
Faculty with Ph.D.s	The proportion of full-time faculty members with a doctorate or the highest degree possible in their field or specialty during the 1997 academic year (3 percent)
Freshman retention rate	Percent of first-year freshman who returned to the same college or university the following fall, averaged over the classes entering between 1993 and 1996 (4 percent)
Full-time faculty	The proportion of total faculty employed on a full-time basis during the 1997 academic year (1 percent)
Graduation rate	Percentage of freshman who graduated within a 6-year period, averaged over the classes entering between 1988 and 1991 (16 percent)
Graduation rate performance	The difference between the actual 6-year graduation rate for students entering in the fall of 1991 and the rate expected from entering test scores and education expenditures. In previous years this criterion was referred to as 'value added' (5 percent)
High school class standing	The proportion of students enrolled for the fall 1997 academic year who graduated in the top 10 percent (for national universities and liberal arts colleges) or 25 percent (for regional universities and liberal arts colleges) of their high school class (5.25 percent)
SAT/ACT scores	Average test scores on the SAT or ACT of enrolled students, converted to percentile scores by using the distribution of all test takers (6 percent)
Student/faculty ratio	The ratio of full-time-equivalent students to full-time equivalent faculty members during the fall 1997 semester, excluding faculty and students of law, medical, and other stand-alone graduate or professional programs in which faculty teach virtually only graduate-level students. Faculty members also exclude graduate teaching assistants (1 percent)
Yield	The ratio of students who enrolled to those admitted to the fall 1997 freshman class (1.5 percent)

USNWR reported data on 14 ranking criteria for tier 1 national universities. Eleven of 14 ranking criteria were reported for tier 3 through tier 4 national universities. For consistency, the analysis presented below was restricted to this subset of ranking criteria. Table 2 summarizes the subset of 11 raking criteria analyzed in this study.

According to the weights provided in Table 1, the 11 ranking criteria analyzed in this study account for 82.5 percent of the USNWR tier rankings. In descending order of importance, these criteria are academic reputation (REP, 25 percent); the 6-year graduation rate (ACTGRAD, 16 percent); average SAT scores (SATAVG, 6 percent); the percentage of classes with enrollment of less than 20 students (LT20, 6 percent); the proportion of students enrolled who graduated in the top 10 percent of their high school class (TOP10, 5.25 percent); the predicted graduation rate (PREDGRAD, 5 percent); the percentage of undergraduate alumni who contributed to the university (ALUM, 5 percent); the retention rate (RET, 4 percent); the undergraduate acceptance rate (ACCRAT, 2.25 percent); the percentage of undergraduate classes with more than 50 students (MT50, 2 percent); and the proportion of total faculty employed on a full-time basis (FTFAC, 1 percent).

4. Empirical analysis

Table 3 summarizes the means and standard deviations of the 11 ranking criteria for the national univer-

Table 2
Definitions of *U.S. News & World Report* explanatory variables

Variable	Definition
ACCRAT (X_1)	The ratio of the number of students admitted to the number of applicants for admission for the fall 1997 semester
ACTGRAD (X_2)	The 6-year graduation rate for students entering in the fall of 1991
ALUM (X_3)	The percentage of undergraduate alumni of record who donated money to the institution during the years 1996 and 1997
FTFAC (X_4)	The proportion of total faculty employed on a full-time basis during the 1997 academic year
LT20 (X_5)	The percentage of undergraduate classes, excluding class sections, with fewer than 20 students enrolled during the fall 1997 semester
MT50 (X_6)	The percentage of undergraduate classes, excluding class sections, with 50 students or more enrolled during the fall 1997 semester
PREDGRAD (X_7)	The predicted graduation rate, which is based on entering test scores and education expenditures
REP (X_8)	The average rating of the quality of a school's academic programs as evaluated by officials at similar institutions. The survey was conducted in the spring of 1998
RET (X_9)	The ratio of the number of students admitted to the number of applicants for admission for the fall 1997 semester
SATAVG (X_{10})	Average of the first and fourth percentile SAT test scores of enrolled students
TOP10 (X_{11})	The proportion of students enrolled in national universities and liberal arts colleges in the fall 1997 academic year who graduated in the top 10 percent of their high school class

Table 3
Comparative simple statistics—all tiers (sample mean, sample standard deviation; n=114)

Variable	Mean	Standard deviation
ACCRAT	62.00	21.61
ACTGRAD	67.92	16.91
ALUM	21.41	12.31
FTFAC	86.46	11.56
LT20	48.75	15.05
MT50	11.75	6.41
PREDGRAD	66.70	15.12
REP	3.38	0.74
RET	85.39	8.85
SATAVG	1189.09	129.55
TOP10	48.07	28.09

sities analyzed in this study for which a complete data set was available. Of 114 institutions examined, for example, the average SAT score (X_{10}) was 1189, the average retention rate was about 85 percent, the average reputation score was 3.4, and so on.

The statistics presented in Table 3 are biased upwards because a greater percentage of the more prestigious (tier 1 and 2) national universities require that applicants submit SAT scores, while a larger proportion of the less prestigious (tier 3 and 4) national universities require that applicants submit ACT scores, which are not included in this study. Comparative simple statistics for each tier group are summarized in Table 4.

A cursory examination of the data presented in Table 4 suggests that a systematic relationship exists between the USNWR tier rankings and the ranking criteria. The higher the average SAT scores, academic reputation scores, retention rates, predicted graduation rates, actual graduation rates, percent of full-time faculty, percentage of classes with enrollment of less than 20 students, etc., for example, the higher the tier ranking (i.e. the lower the tier number). The data presented in the table also indicate conflicting relationships. The average percentage of classes with enrollment of greater than 50 students for tier 1 universities, for example, is 11.4 percent, while for tier 2 universities it is 12.7 percent. Yet, for tier 3 and 4 universities the percentage of classes with enrollment of greater than 50 students is 9.5 percent and 10.3 percent, respectively.

The relationship between tier rankings and the MT50 is controversial since a positive weight is assigned to this ranking criteria. From the student's perspective it could be argued on intuitive grounds that smaller classes are preferable to larger classes. This would imply that a negative weight should be applied to this ranking criteria. On the other hand, it could also be argued that large class size allows institutions to allocate their financial resources more efficiently, which would justify a positive weight.

A similar observation may be made about the ratio of the number of students admitted to the number of admis-

Table 4
USNWR simple statistics

Variable	*n*	Mean	Std. dev.	*n*	Mean	Std. dev.
	Tier 1			**Tier 2**		
ACCRAT	46	42.13	18.50	42	73.24	10.55
ACTGRAD	46	83.24	8.33	42	65.43	7.00
ALUM	46	31.24	12.75	42	17.12	6.67
FTFAC	46	90.74	6.80	42	86.62	13.19
LT20	46	57.13	12.44	42	44.74	6.87
MT50	46	11.41	5.49	42	12.71	6.87
PREDGRAD	46	81.30	10.21	41	61.59	6.63
REP	46	4.08	0.54	42	3.15	0.37
RET	46	93.11	3.35	42	84.45	3.70
SATAVG	46	1315.63	91.24	42	1141.10	53.31
TOP 10	44	76.57	16.46	41	35.34	16.72
	Tier 3			**Tier 4**		
ACCRAT	21	77.21	10.71	22	74.68	10.63
ACTGRAD	21	51.95	11.11	22	38.59	11.34
ALUM	21	12.81	4.82	22	10.86	6.97
FTFAC	21	78.90	12.46	22	80.86	13.60
LT20	21	43.48	7.67	22	38.00	17.02
MT50	21	9.47	6.23	22	10.27	7.31
PREDGRAD	20	53.70	7.70	19	46.53	8.04
REP	21	2.74	0.22	22	2.39	0.32
RET	21	77.29	7.45	22	21.36	5.48
SATAVG	21	1077.24	62.82	22	1024.36	64.30
TOP 10	20	25.65	16.29	19	21.15	9.25

sion applications on the grounds that the more prestigious the institution, the lower should be the acceptance rate. An examination of Table 4 suggests, however, that this is not necessarily the case. The acceptance rate for tier 4 universities (74 percent), for example, is less than the acceptance rate for tier 3 universities (77 percent). Although this difference is not statistically significant at traditional confidence levels at the indicated standard deviations, it does underscore a conceptual difficulty similar to that associated with MT50.

The relationship between acceptance rates and tier rankings appears to ignore the related phenomenon of applicant self-selection. Self-selection refers to the situation in which academically under-qualified students do not apply to tier 1 institutions because, in their judgment, their applications would be rejected. Because of the transaction costs involved in the application process (application fees, time, etc.), the effect of self-selection would be to exert an upward bias on reported acceptance rates for more prestigious institutions. Moreover, as the number of applications by academically under qualified students to institutions of lower prestige increase, acceptance rates at these national universities are biased downwards.

To further examine the statistical relationship between USNWR tier rankings, ordinary-least-squares regression analysis was performed on the 11 ranking criteria summarized in Table 4.[1] The regression results indicate that variations in the complete set of ranking criteria explain about 81 percent of the total variation in tier rankings. This is consistent with the 82.5 percent contribution of the 11 USNWR ranking criteria considered. The unrestricted *F*-statistic suggests that the regression equation is statistically significant at greater than the 99 percent confidence level. The remaining regression statistics are less encouraging.

Of the 11 ranking criteria examined, six are apparently statistically insignificant at the 95 percent confidence level for a one-tailed test. This result is noteworthy since the USNWR tier index is arbitrarily constructed using preselected ranking criteria. Moreover, the ranking for three of the criteria (ALUM, SATAVG, and TOP10) are incorrectly signed, which would suggest the possibility of multicollinearity. To investigate this possibility, pairwise correlation coefficients were estimated.[2]

Estimated pairwise correlation coefficients indicate

[1] Detailed statistical results available from the author on request.

[2] Multicollinearity is the condition where one or more of the ranking criteria are interrelated. The main consequences of multicollinearity are unreliable *t*- and *F*-statistics, and incorrectly signed parameter estimates.

that variations in tier rankings (TIER) are most closely associated with variations in ACTGRAD (−0.86), RET (−0.84), REP (−0.80), SATAVG (−0.78), TOP10 (−0.74), and ACCRAT (0.67). This is significant since the regression results suggest that SATAVG, TOP10, and ACCRAT were statistically insignificant. Moreover, the nature of the relationship between ACCRAT and TIER conforms to theory, i.e. a lower acceptance rate is likely to be associated with a more prestigious institution.

Perhaps the most striking aspect of the estimated correlation matrix presented is the pervasiveness and severity of multicollinearity in the data. Average SAT scores (SATAVG) of enrolled students and the predicted graduate rate (PREDGRAD), for example, are nearly perfectly correlated (0.96). Retention rates (RET) and the actual graduation rates (ACTGRAD) are also highly correlated (0.90). Similar results are evident for PREDGRAD and ACTGRAD (0.84), the acceptance rate (ACCRAT) and PREDGRAD (−0.84), SATAVG and ACCRAT (−0.85), SATAVG and ACTGRAD (0.81), academic reputation (REP) and ACTGRAD (0.80), alumni contributions (ALUM) and ACTGRAD (0.72), ALUM and PREDGRAD (0.75), ALUM and SATAVG (0.77), REP and PREDGRAD (0.80), RET and PREDGRAD (0.81), REP and ACCRAT (−0.78), REP and RET (0.76), REP and SATAVG (0.82), REP and TOP10 (0.77), RET and SATAVG (0.80), etc.

A further review of the estimated correlation coefficient matrix reveals that the highest degree pairwise correlations (coefficients greater than 0.70) involve academically related ranking criteria, such as acceptance rates, graduation rates, predicted graduation rates, academic reputation, retention rates, SAT scores, and the proportion of students graduating in the top 10 percent of their high school class. Of 24 pairwise correlations that fall into this category, seven involved average SAT scores, seven involved predicted graduation rates, seven involved actual graduation rates, six involved academic reputation, six involved retention rates, six involved acceptance rates, and three involved alumni contributions. The only ranking criteria that were not significantly correlated were non-academically related factors, such as class size and percentage of full-time faculty.

The estimated correlation coefficients also suggest that the relative weights assigned by USNWR to the ranking criteria may not be representative of their actual contributions to TIER. A university's academic reputation, for example, is highly correlated with the academic quality of its students, as measured by average entering SAT scores. Academic reputation is also highly correlated with undergraduate retention rates, which may indicate that students who believe that the institution's degree is highly valued in the marketplace are more likely to remain with that institution. On the other hand, students who come to believe that the institution's degree is tainted by an inferior academic reputation may in time transfer to a more prestigious college or university. A possible chain of causality may be from admission standards to academic quality to retention. Improving retention rates, therefore, may depend, in part, on higher admission standards.

Academic reputation is also positively correlated with alumni contributions. Alumni who are proud of their alma mater and who believe that their degree provides a competitive advantage in the marketplace may be more prone to provide financial assistance after graduation than those who believe that they did not get their tuition's worth. The positive correlation between academic reputation and alumni contributions is underscored by the fact that major corporations tend to allocate their scarce recruitment dollars to institutions with high academic reputations (tiers 1 and 2), and tend to shun those colleges and universities perceived to be inferior (tiers 3 and 4). The relationship between academic reputation and alumni contributions is particularly important, since a university's ability to provide state-of-the-art educational facilities and hire world-class faculty depend crucially on non-tuition sources of income. As above, a possible chain of causality may be from admission standards to academic quality to alumni contribution, Raising alumni contributions, therefore, may partly depend on higher admission standards.

The presence of severe and pervasive multicollinearity suggests that the weights assigned by USNWR may not accurately reflect the actual contributions of the ranking criteria to an institution's overall tier ranking. It could be argued that an accurate understanding of the actual contribution of the ranking criteria to an institution's overall academic standing is an essential element in the formulation of a university's strategic agenda.

The following discussion of principal component analysis explains why this methodology was used to further explore the relative importance of individual explanatory variables in the presence of severe multicollinearity.

4.1. Principal component analysis

Although parameter estimates derived from ordinary least squares in the presence of multicollinearity are BLUE (best linear unbiased estimates), they can no longer be interpreted as first partial derivatives. Thus, variations in one explanatory are accompanied by variations in one or more of the remaining explanatory variables in the model. In other words, it is difficult to isolate the partial effect of individual explanatory variables on the dependent variable (see, for example, Ramanathan, 1998). This is important since we would like to know whether the arbitrary weighting scheme adopted by *U.S. News & World Report* to determine its tier rankings is an accurate reflection of the contribution of each of the ranking criteria.

One solution to the multicollinearity problem suggested in the literature is principal component regression analysis (see, for example Chatterjee & Price, 1977; Green, 1997; Hair, Anderson & Tatham, 1987; Hotelling 1933, 1936; Judge, Griffiths, Hill & Lee, 1985; Maddala, 1992; Malinvaud, 1997). The objective of principal component analysis is to derive an alternative set of explanatory variables, called principal components, that have certain desirable statistical properties.

The strategy underlying the principal component methodology is straightforward. Suppose that we have identified k explanatory variables in a functional relationship. Consider the following linear functions of these variables:

$$z_1=a_1x_1+a_2x_2+\ldots+a_kx_k \tag{1.1}$$

$$z_2=b_1x_1+b_2x_2+\ldots+b_kx_k \tag{1.2}$$

$$\vdots$$

$$z_k=k_1x_1+k_2x_2+\ldots+k_kx_k \tag{1.k}$$

Suppose that we choose values of as in Eq. (1.1) such that variances are maximized subject to the requirement that

$$a_1^2+a_2^2+\ldots+a_k^2=1 \tag{2.1}$$

This restriction, which is referred to as the normalization condition, is necessary, otherwise the value of z_1 could be indefinitely increased. z_1 is called the first principal component. It is a linear function of the explanatory variables (x_i) that has the greatest variance. Intuitively, the first principal component should be able to explain variations in the value of the dependent variable better than any other linear combination of explanatory variables subject to the normalization rule. We may then consider other linear functions, such as z_2 which is uncorrelated with z_1, subject to the condition that

$$b_1^2+b_2^2+\ldots+b_k^2=1 \tag{2.2}$$

z_2 is said to be the second principal component. This procedure is repeated until we have k linear functions z_1, $z_2,\ldots,z_k$, which are called the principal component of the x_is. The variances of the principal component may be ordered as

$$var(z_1)>var(z_2)>\ldots>var(z_k) \tag{3}$$

Each principal component is a linear combination of the original variables, with coefficients equal to the eigenvectors of the correlation or covariance matrix. The normalization condition guarantees that the eigenvectors have unit length. The principal components are then sorted in descending order by eigenvalue, which are equal to the variances of the components.

In geometric terms, principal component analysis is similar to the method of ordinary least squares. Principal component analysis of the k-dimensional linear subspace spanned by the first k principal component gives the best possible fit to the data points as measured by the sum of squared perpendicular distances from each data point to the surface. Ordinary least squares regression analysis, on the other hand, minimizes the sum of the squared vertical distances.

Principal components have certain desirable properties. The first is that the sum of the variances of the principal component is equal to the sum of the variances of the original explanatory variables, that is

$$var(z_1)+var(z_2)+\ldots+var(z_k)=var(x_1)+var(x_2)+\ldots+var(x_k) \tag{4}$$

The second is that, unlike the original explanatory variables, the z_is are mutually orthogonal (that is, they are uncorrelated). In other words, there is zero multicollinearity among the z_is.

Although principal component analysis appears to offer a solution to the multicollinearity problem, the procedure has a few drawbacks. To begin with, although the first principal component has the greatest variance subject to the normalization condition, it need not necessarily be the most highly correlated with the dependent variable. In other words, there is no necessary relationship between the order of the principal components and their degree of correlation with the dependent variables.

Another drawback with this methodology is that the z_is have no meaningful economic interpretation. Finally, changing the units of measurement of the x_is will change the principal component. To overcome this problem, all of the ranking criteria considered in this study have been standardized to have unit variance.

Table 5 summarizes the eigenvalues of the 11 calculated principal component and their proportional explanatory contributions applied to the *U.S. News & World Report* tier rankings. The eigenvalues indicate that the first principal component (Prin1) explains about 62 percent of the standardized variance in TIER, the second principal component (Prin2) explains another 18 percent, the third principal component (Prin3), another 6 percent, and so on.

The first four principal components, which explain about 79 percent of the standardized variance in TIER, are summarized in Table 6. The first principal component, which explains variations in the value of the dependent variable better than any other linear combination of explanatory variables, is a measure of the overall USNWR tier rankings. The first eigenvector reveals that there are approximately equal loadings on eight of the 11 ranking criteria examined, accounting for approximately 88 percent of the absolute standardized variance. These eight ranking criteria, in descending order of their

Table 5
Principal component analysis. Eigenvalues,[3] proportion explained, and cumulative total (n=114)

Principal component	Eigenvalue	Difference	Proportion explained	Cumulative total
Prin1	6.810	4.886	0.619	0.619
Prin2	1.924	1.320	0.175	0.794
Prin3	0.603	0.719	0.055	0.849
Prin4	0.425	0.060	0.039	0.888
Prin5	0.365	0.106	0.033	0.921
Prin6	0.259	0.058	0.024	0.944
Prin7	0.201	0.045	0.018	0.963
Prin8	0.156	0.006	0.014	0.977
Prin9	0.150	0.081	0.014	0.990
Prin10	0.070	0.033	0.006	0.997
Prin11	0.036	–	0.003	1.000

Table 6
Principal component analysis. Eigenvalues (n=113)

Variable	Prin1	Prin2	Prin3	Prin4
ACCRAT	−0.335	0.055	0.234	0.081
ACTGRAD	0.350	0.047	−0.006	−0.231
ALUM	0.307	−0.058	0.499	−0.459
FTFAC	0.132	0.524	0.677	0.309
LT20	0.216	−0.483	0.116	0.614
MT50	−0.020	0.654	−0.362	0.100
PREDGRAD	0.364	−0.095	−0.077	−0.049
REP	0.341	0.143	−0.076	0.138
RET	0.337	0.089	−0.197	−0.266
SATAVG	0.366	−0.049	−0.019	0.022
TOP10	0.330	0.125	−0.205	0.392

individual contribution to the standardized variance of the first principal component, are SATAVG (11.8 percent), PREDGRAD (11.7 percent), ACTGRAD (11.3 percent), REP (11 percent), RET (10.9 percent), ACCRAT (10.8 percent), TOP10 (10.6 percent), and ALUM (9.9 percent).

Not surprisingly, these eight ranking criteria in Table 6 had the highest pairwise correlation coefficients with TIER. The second eigenvector has high positive loadings on MT50 and FTFAC, and a high negative loading on LT20. The interpretation of the second principal component for MT50 and LT20 is fairly straightforward. Although its overall contribution to TIER is relatively minor, the high positive loading on FTFAC appears counterintuitive. It surprisingly suggests that the greater the percentage of full-time faculty, the lower (higher number) an institution's tier ranking. An interpretation of the remaining principal components is not obvious.

A simple way to assess the explanatory power of principal component regression analysis is to simulate the predictions of the first principal component, which explains variations in the value of the dependent variable better than any other linear combination of explanatory variables subject to the normalization rule.[1] Although there is a small amount of overlap of USNWR tier rankings when compared with re-centered first principal component estimates, the simulation underscores the explanatory power of the principal component methodology. Re-centered first principal component estimates of tier 1 national universities, for example, range in value from 0.25 to approximately 2.5. Tier 2 national universities range in value from about 2.4 to around 3.3. First principal component estimates of tier 3 institutions range from about 3.2 to about 3.7. Lastly, first principal component estimates of tier 4 universities range from about 3.2 to around 4.2.

[3] Eigenvalues represent the column sum of squares for a factor, sometimes referred to as a latent root. It represents the amount of variance accounted for by a factor.

5. Conclusions

This paper utilized principal component regression analysis to examine the relative contributions of 11 ranking criteria used to construct USNWR tier rankings of national universities. The main finding of this study is that the actual contributions of the 11 ranking criteria examined differ significantly from the explicit USNWR weighting scheme because of the presence of severe and pervasive multicollinearity. The eigenvalues of the first principal component presented in Table 6 indicate that the importance of ranking criteria in explaining the USNWR tier rankings, in descending order (the USNWR ranking order is in parentheses), are SATAVG (3), PREDGRAD (6), ACTGRAD (2), REP (1), RET (8), ACCRAT (9), TOP10 (5), ALUM (7), LT20 (4), FTFAC (11), and MT50 (10).

Principal component analysis of the 11 ranking criteria examined in this paper indicates that academic reputation, which is the most heavily weighted USNWR ranking criterion (25 percent), ranked fourth (11 percent) when the effects of multicollinearity were explicitly considered. The eigenvalue of the first principal component indicates, on the other hand, that the most significant ranking criterion (11.7 percent) was average SAT scores of enrolled students, which ranked a distant third (6 percent) according to the USNWR weighting scheme. The importance of SATAVG in explaining the USNWR tier stems not only from its direct effect on tier rankings but also from its indirect effect on seven of the remaining ten ranking criteria, including (in descending order) actual graduation rates, predicted graduation rates, retention rates, alumni contributions, academic reputation, the percentage of enrolled students who graduated in the top 10 percent of their high school class, and the acceptance rate.

6. Suggestions for further research

An obvious avenue for further research relates to the data set employed in this study. To begin with, the analysis should be expanded to include colleges and universities that require ACT scores. In fact, standardized conversion tables are available that allow admissions officers of colleges and universities to easily convert ACT scores to comparable SAT scores. Another extension of this research is to conduct similar analyses on national liberal arts colleges and regional universities and liberal arts colleges, collectively and regionally.

Other avenues of further research are more subtle. The results presented in this paper suggest that a policy of lower admission standards to bolster enrollments to increase or maintain tuition-based revenues may be counterproductive. Lowering SAT entrance requirements may lead to a decline in academic reputation and tier rankings, which could lead to a decline in admission applications by better academically qualified students. This, in turn, could contribute to a decline in lower enrollments and declines in tuition-based revenues. Further lowering SAT entrance requirements to compensate could result in a 'vicious cycle' where academic reputation, enrollments, and revenues are further eroded.

The above discussion suggests that further research on the interrelationship between SAT scores, academic reputation, and retention rates is necessary. It could be argued, for example, that SAT scores influence retention rates, with academic reputation the transmission mechanism. Moreover, this transmission mechanism may operate at several levels. A university's degree has an imputed market value. Investors in university degrees will at least implicitly compare the marginal cost of a college education with the marginal benefits associated with increased career opportunities, lifetime earnings, etc. If the marginal cost of a degree is greater than the perceived marginal benefits, then students are likely to seek admission to another college or university where the cost–benefit relationship is more favorable. The student will do this by transferring to a university with a better academic reputation, or an institution where tuition costs are lower, or both.

Another area of further research is the relationship between retention rates and classroom management. It could be argued that retention rates decline when students with above average academic qualifications become dissatisfied with a classroom environment that accommodates academically lesser qualified students. A lower level and slower pace of instruction may result in academically superior students transferring to what they perceive to be academically superior institutions where instruction is more advanced and accelerated. This phenomenon might be described as Gresham's law of higher education with 'bad' students driving out 'good' students.[4] Moreover, efforts to bolster enrollments by increasing the size of the freshman class by lowering admission standards could be more than offset by the loss of sophomores and juniors transferring to other institutions.

Another area for further research would focus on the observation that SAT admission requirements for entering freshmen are higher than for transfer students. It could be argued that students use their second- and

[4] 'Gresham's Law', as it is generally understood, states that 'bad' money tends to drive 'good' money out of circulation. The distinction involves two or more species, say, gold and silver, of different intrinsic value but with the same face value. According to this 'law', undervalued coins, i.e. coins with a face value less than their intrinsic value (say, gold), will tend to disappear from circulation, while overvalued coins (say, silver) continue to be used for transaction purposes.

third-choice colleges and universities as stepping stones to admission into their first- and second-choice colleges and universities after the first year.

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